

Submission A solution to Problem 9

Hook coefficients in the universal Hilbert series of diagonal coinvariants

Problem statement and interpretation

I interpret $R_n^{(m)}$ as the diagonal coinvariant algebra

$$R_n^{(m)} = \mathbb{C}[x_i^{(r)} : 1 \leq i \leq n, 1 \leq r \leq m] / \langle \mathbb{C}[x_i^{(r)}]_+^{\mathfrak{S}_n} \rangle,$$

where $\mathfrak{S}_n$ acts by simultaneously permuting the subscript i in all m sets of variables. Thus the coefficients $c_\lambda(n)$ are the stable multiplicities of the polynomial GL_m -irreducibles $S_\lambda(\mathbb{C}^m)$.

The combinatorial model

For $r, k \geq 1$, let $\text{OP}_{r,k}$ be the set of ordered set partitions $\sigma = (B_1 | \cdots | B_k)$ of $[r]$ into k nonempty blocks. An inversion of σ is a pair (i, j) such that

$$i < j, \quad i = \min B_m, \quad j \in B_\ell, \quad \ell < m.$$

Let $\text{inv}(\sigma)$ be the number of such pairs. When $k = r$, this is the usual inversion number of the permutation obtained by reading the singleton blocks from left to right.

Theorem 1. *Let $\lambda = (a, 1^b)$ be a hook, with $a \geq 1$ and $b \geq 0$. Then $c_{(a, 1^b)}(n)$ is the number of ordered set partitions $\sigma = (B_1 | \cdots | B_k)$ of $[r]$, for some $2 \leq r \leq n$, such that*

$$r - k = b + 1, \quad r \text{ is not a singleton block of } \sigma, \quad \text{inv}(\sigma) = a - 1.$$

Equivalently,

$$\sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b = \sum_{\substack{2 \leq r \leq n \\ 1 \leq k \leq r-1}} \sum_{\substack{\sigma \in \text{OP}_{r,k} \\ r \text{ not singleton in } \sigma}} q^{\text{inv}(\sigma)} z^{r-k-1}.$$

The empty partition coefficient is $c_\emptyset(n) = 1$.

Proof. We use two standard inputs. First, diagonal supersymmetry says that the universal Schur coefficients for diagonal coinvariants can be read from the coinvariant ring with one bosonic alphabet and one fermionic alphabet,

$$SR_n = (\mathbb{C}[x_1, \dots, x_n] \otimes \bigwedge(\theta_1, \dots, \theta_n)) / \langle (\mathbb{C}[x] \otimes \bigwedge \theta)_+^{\mathfrak{S}_n} \rangle.$$

More explicitly, the supersymmetric Frobenius/character expansion has the same universal multiplicity modules as the purely bosonic expansion; after applying the dimension map to the $\mathfrak{S}_n$ -Frobenius factor one obtains exactly the Hilbert coefficients $c_\lambda(n)$ appearing in the question. Therefore

$$\text{Hilb}(SR_n; q, z) = \sum_{\lambda} c_\lambda(n) s_\lambda(q/z),$$

where $s_\lambda(q/z)$ denotes the super Schur polynomial in one even variable q and one odd variable z . With one even and one odd variable, $s_\lambda(q/z)$ vanishes unless λ lies in the $(1, 1)$ -hook, i.e. unless λ is a hook, and

$$s_{(a, 1^b)}(q/z) = q^a z^b + q^{a-1} z^{b+1} = q^{a-1} z^b (q + z).$$

Thus

$$\text{Hilb}(SR_n; q, z) = 1 + (q + z) \sum_{a \geq 1, b \geq 0} c_{(a, 1^b)}(n) q^{a-1} z^b. \quad (1)$$

Second, the Hilbert series of SR_n , due to Rhoades–Wilson, is

$$H_n(q, z) := \text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} [k]!_q \text{Stir}_q(n, k), \quad (2)$$

where $[k]_q = 1 + q + \dots + q^{k-1}$, $[k]!_q = [k]_q [k-1]_q \dots [1]_q$, and

$$\text{Stir}_q(r, k) = \text{Stir}_q(r-1, k-1) + [k]_q \text{Stir}_q(r-1, k).$$

The boundary conditions are $\text{Stir}_q(0, 0) = 1$, $\text{Stir}_q(r, 0) = 0$ for $r > 0$, and $\text{Stir}_q(r, k) = 0$ outside $0 \leq k \leq r$. Put $F_{r,k} = [k]!_q \text{Stir}_q(r, k)$. Then $F_{r,k} = 0$ outside $1 \leq k \leq r$, except for the harmless convention $F_{0,0} = 1$, and

$$F_{r,k} = [k]_q F_{r-1, k-1} + [k]_q F_{r-1, k}.$$

Using $[k+1]_q = 1 + q[k]_q$ in (2), we get, for $n \geq 2$,

$$\begin{aligned} H_n(q, z) &= \sum_{k=1}^n z^{n-k} ([k]_q F_{n-1, k-1} + [k]_q F_{n-1, k}) \\ &= H_{n-1}(q, z) + (q + z) \sum_{k=1}^{n-1} [k]_q F_{n-1, k} z^{n-1-k}. \end{aligned}$$

Since $H_1(q, z) = 1$, iteration gives

$$\frac{H_n(q, z) - 1}{q + z} = \sum_{r=1}^{n-1} \sum_{k=1}^r [k]_q F_{r,k} z^{r-k}. \quad (3)$$

It remains to identify the right side of (3). The polynomial $F_{r,k}$ is the inversion enumerator of ordered set partitions:

$$F_{r,k} = \sum_{\tau \in \text{OP}_{r,k}} q^{\text{inv}(\tau)}. \quad (4)$$

Indeed, both sides satisfy the same recurrence. To see this directly, insert the largest letter r into an ordered partition of $[r-1]$. To obtain k final blocks, either insert r as a singleton block into one of the k possible final block positions, or insert r into one of the k existing blocks. If the final position is i from the left, the new inversions involving r are exactly the minima of the blocks to its right, so their number is $k-i$. Summing over $i = 1, \dots, k$ contributes $[k]_q$ in each of the two cases, giving the recurrence above and the initial condition.

Now restrict this insertion argument to insertions of $r+1$ into an existing block of an ordered partition $\tau \in \text{OP}_{r,k}$. The result is precisely an ordered partition $\sigma \in \text{OP}_{r+1,k}$ in which $r+1$ is not a singleton, and the extra inversion contribution is again one of $0, 1, \dots, k-1$. Hence

$$[k]_q F_{r,k} = \sum_{\substack{\sigma \in \text{OP}_{r+1,k} \\ r+1 \text{ not singleton}}} q^{\text{inv}(\sigma)}.$$

Substituting this into (3), and replacing $r+1$ by r , gives exactly

$$\sum_{a \geq 1, b \geq 0} c_{(a,1^b)}(n) q^{a-1} z^b = \sum_{\substack{2 \leq r \leq n \\ 1 \leq k \leq r-1}} \sum_{\substack{\sigma \in \text{OP}_{r,k} \\ r \text{ not singleton}}} q^{\text{inv}(\sigma)} z^{r-k-1}.$$

Extracting the coefficient of $q^{a-1} z^b$ proves the claimed interpretation. $\square$

References

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- [2] J. Lentfer, *Diagonal supersymmetry for coinvariant rings*, arXiv:2505.14885 (2025).
- [3] B. Rhoades and A. T. Wilson, *The Hilbert series of the superspace coinvariant ring*, Forum Math. Pi 12 (2024), e16.